DBMS

Assignment No.-4

Date

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B.Tech CSE Core - Section-A

Section - A :-

1. Transaction :-
- A logical unit of work consisting of multiple operations.
- Example : Bank transfer (debit + credit)

2. ACID Properties :-
- Atomicity : All or nothing.
- Consistency : Database remains valid.
- Isolation : Transactions don't interfere
- Durability : Changes are permanent.

3. Concurrency Control :-
- Manages simultaneous transactions without conflict.
- Ensures data consistency

4. Deadlock :-
- when transactions wait indefinitely for each other
- Example : T1 waits for T2
  T2 waits for T1

5. Serializability :-
- Concurrent execution gives same result as serial execution.

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6. Recovery in DBMS :-
- Restores database after failure / crash

7. Checkpoint :-
- Saves database state to reduce recovery time.

Section - B :-

1. ACID (Banking Example) :-
- Atomicity :-
  Transfer fully happens or not at all
- Consistency :-
  Total balance remains same
- Isolation :-
  No interference from other transactions
- Durability :-
  Saved even after crash

2. Deadlock Example :-
- T1 locks Account A → wants B.
- T2 locks Account B → wants A.
  Both wait → Deadlock

3. Serial vs Concurrent :-
- Serial : One after another (safe but slow)
- Concurrent : Multiple at same time (fast but needs control).

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Section - C :-

1. Banking Transfer + ACID :-
- Atomicity → Debit & credit both complete
- Consistency → Balance correct.
- Isolation → No conflict
- Durability → Saved permanently.

2. Two Transactions Same data :-
⇒ Issues :-
- Lost update
- Dirty read.
- Inconsistent data

3. System Crash (Recovery steps) :-
- use logs.
- Perform rollback (undo) or redo.
- Restore last checkpoint.

Section - D :-

1. ACID Example (Atomicity) :-
- ATM withdrawal : If cash not dispended - no money deducted.

2. Transaction Timeline (Conflict) :-
⇒ Example :-
- T1 reads A

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- T2 updates A.
- T1 updates A.

Conflict → Inconsistent result.

3. Recovery Technique :-
- Log-based recovery :-
- Undo incomplete transactions.
- Redo completed ones